

PROJECT PRESENTATION

Silver Commodity Price Forecasting

and Trading Analytics System

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**FORECAST
ANALYZE
DECIDE**

ARIMA • XGBoost • LSTM • Ensemble •
Power BI

Problem Statement

Why forecasting needs to be paired with analytics and decision support.

Silver prices respond to market, macroeconomic, and sentiment-driven changes — making timely interpretation difficult.

- High volatility complicates price estimation and investment timing.
- Multiple variables influence movements in the silver commodity market.
- Standalone forecasts are less useful without clear risk and performance insights.
- The project combines forecasting, visualization, and trading simulation in one workflow.

PROJECT OBJECTIVE

**Build a forecasting
and analytics
platform for silver
price prediction
and trading
support.**

Dataset & Features

Historical silver market data covering more than 25 years.

Market data

Open, High, Low, Close

VWAP and futures prices

Historical price trends

Contextual indicators

CPI and interest rates

Supply-demand indicators

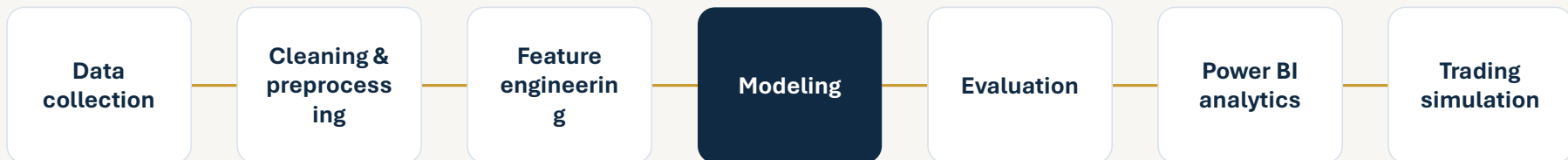
Market sentiment variables

FEATURE

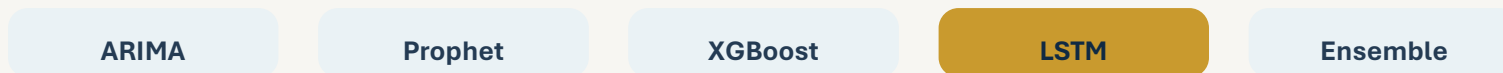
Combined feature set supports both statistical time-series forecasting and machine-learning models.

Methodology

An end-to-end pipeline from raw data to trading insights.



Models evaluated



Model Performance Comparison

Error metrics and goodness-of-fit measured on the evaluation data.

Model	MAE	RMSE	R ²
ARIMA	6.24	7.48	-1.790
XGBoost	1.76	2.30	0.742
LSTM	1.23	1.57	0.886
Ensemble	1.04	1.31	0.835

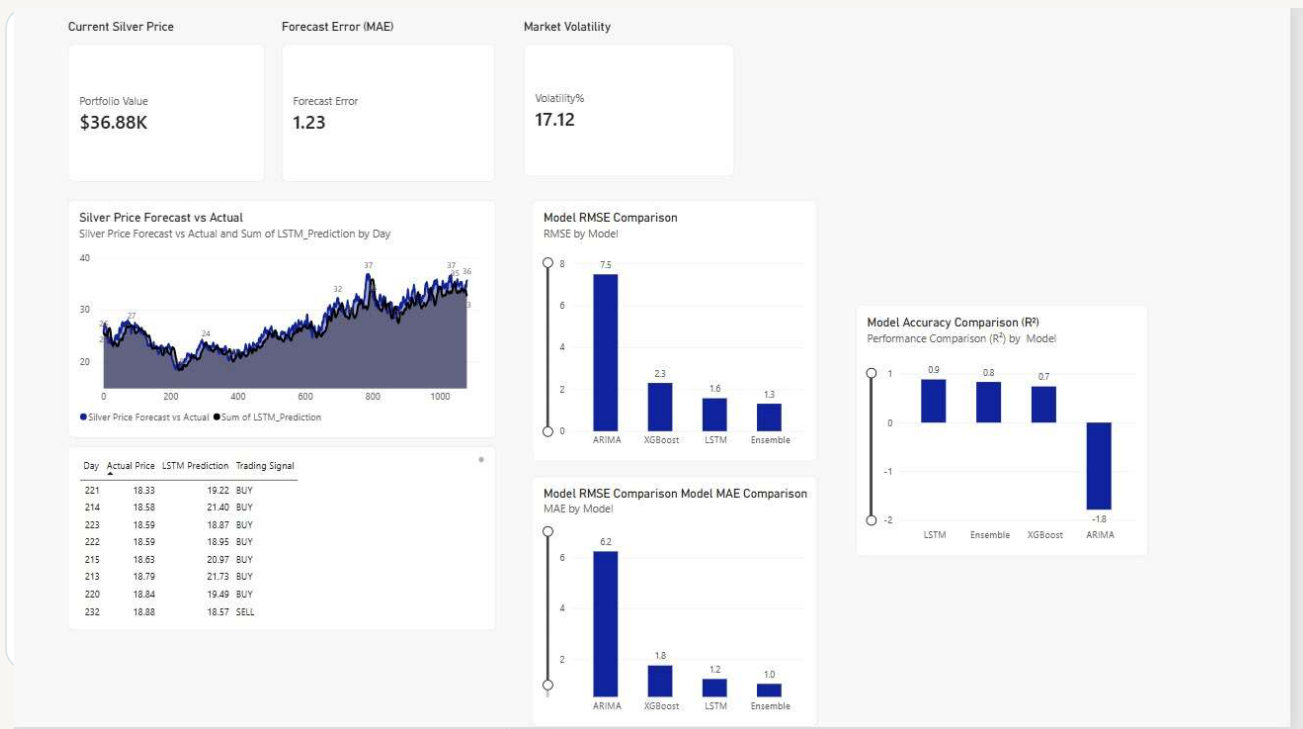
RMSE comparison



Key observation: LSTM achieved the highest R² (0.886), while the ensemble recorded the lowest MAE and RMSE.

Forecast Analytics Dashboard

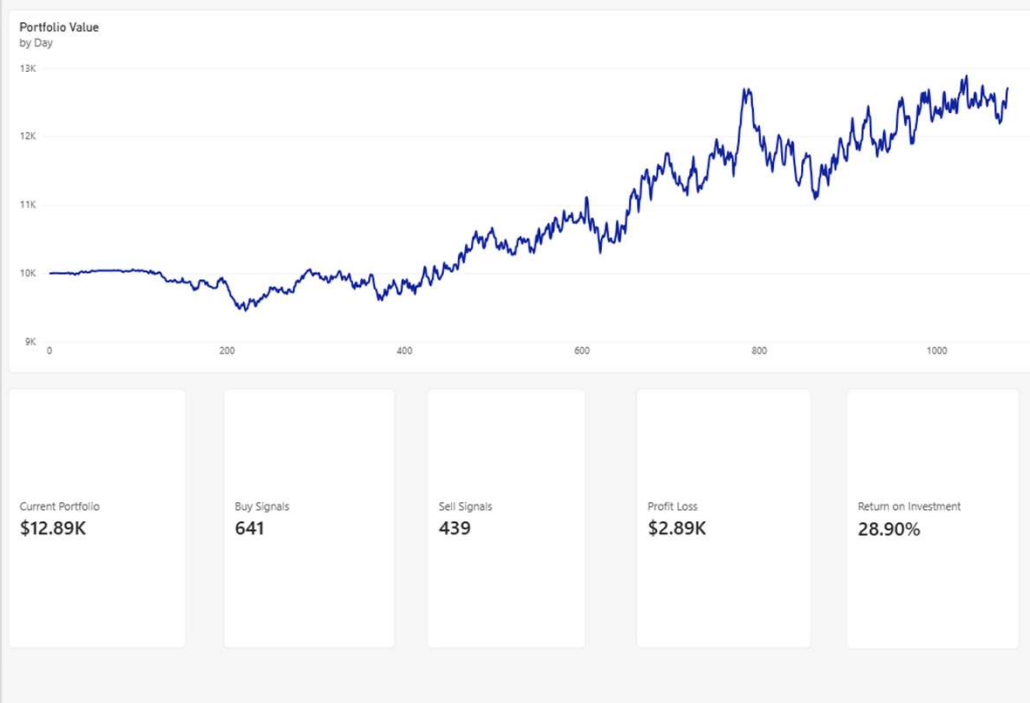
Power BI view for interpreting forecasts, error patterns, and market movement.



- Actual vs forecast visualization
- Forecast error monitoring
- Market volatility analysis
- Trading signal generation
- Model comparison metrics

Commodity Trading Arena

Power BI view for strategy simulation and portfolio performance tracking.



- AI-driven trading simulation
- Portfolio value tracking
- Buy / sell signal monitoring
- Profit and ROI evaluation
- Strategy performance analysis

Key Results

Model outcomes translated into practical forecasting and trading insights.

BEST R^2

0.886

LOWEST MAE

1.04

LOWEST RMSE

1.31

DASHBOARDS

2

- LSTM delivered the strongest goodness-of-fit among the compared models.
- The ensemble reduced MAE and RMSE, supporting more accurate point forecasts.
- Power BI dashboards made forecast behaviour and risk indicators easier to interpret.
- Trading signals enabled a simulated portfolio workflow for decision support.

Future Scope

Possible extensions to improve real-world usefulness and scalability.

01

**Real-time market data
integration**

02

**Advanced ensemble
forecasting**

03

**Automated trading
strategies**

04

**Web and mobile dashboard
deployment**

05

Multi-commodity support